CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

CSPIT — Department of Computer Engineering · ITUE203 Web Development Frameworks

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Practical 3: Responsive UI Design using CSS Grid, Flexbox, and Modern SaaS Theme

Practical Aim & Tools

Aim: Design a responsive modern UI using CSS Grid, Flexbox, media queries, CSS variables, and modern SaaS dashboard layout (Acade UI) for mobile, tablet, and desktop screens.

Tools & Technologies: CSS3, CSS Grid, Flexbox, CSS Custom Properties, Responsive Media Queries

1. Design System & CSS Custom Properties

StudentHub utilizes a modern SaaS design system based on a violet/lavender color scheme (`#6C5CE7` primary brand, `#EEECFE` subtle tinted backgrounds, `#2D3436` high-contrast slate text). All design tokens are declared as CSS Custom Properties (`:root`) allowing seamless dynamic dark mode switching.

2. Responsive App Shell & Grid Layouts

• Desktop Layout (≥ 1024px): 2-column flex app shell with a fixed 240px sidebar and fluid `app-main` viewport.
• Timetable Cards: CSS Grid (`grid-template-columns: repeat(auto-fit, minmax(220px, 1fr))`) rendering class schedule cards with lavender badges.
• Circular SVG Progress Rings: Donut cards displaying 92% Attendance and 85% Activity completion with pure CSS stroke-dashoffset animations.
• Mobile Layout (< 768px): The sidebar collapses into an off-canvas drawer controlled via the topbar hamburger button, switching grids into 1-column stacks.

3. Implementation Code: CSS Tokens & Responsive Shell

Listing: Core Responsive CSS Grid & SaaS Design Tokens

```
:root {
  --primary: #6C5CE7;
  --primary-hover: #5A49E0;
  --primary-light: #EEECFE;
  --bg: #F7F7FD;
```

```
  --surface: #FFFFFF;
  --text: #2D3436;
  --muted: #8395A7;
  --border: #E8E8F3;
  --radius-lg: 18px;
  --shadow-sm: 0 4px 16px rgba(108, 92, 231, 0.06);
}

.app-layout { display: flex; min-height: 100vh; background: var(--bg); }
.app-sidebar { width: 240px; background: var(--surface); border-right: 1px solid var(--border); }
.app-main { flex: 1; display: flex; flex-direction: column; min-width: 0; }
.app-topbar { height: 70px; background: var(--surface); display: flex; align-items: center;
justify-content: space-between; padding: 0 2rem; }

/* Responsive Media Queries */
@media (max-width: 1024px) {
  .app-sidebar { position: fixed; left: -260px; z-index: 1000; transition: left 0.3s ease; }
  .app-sidebar.open { left: 0; }
}
@media (max-width: 768px) {
  .grid.cols-3 { grid-template-columns: 1fr; }
  .app-topbar { padding: 0 1rem; }
}
```

4. Verification Screenshot: Responsive Layout & About Credentials

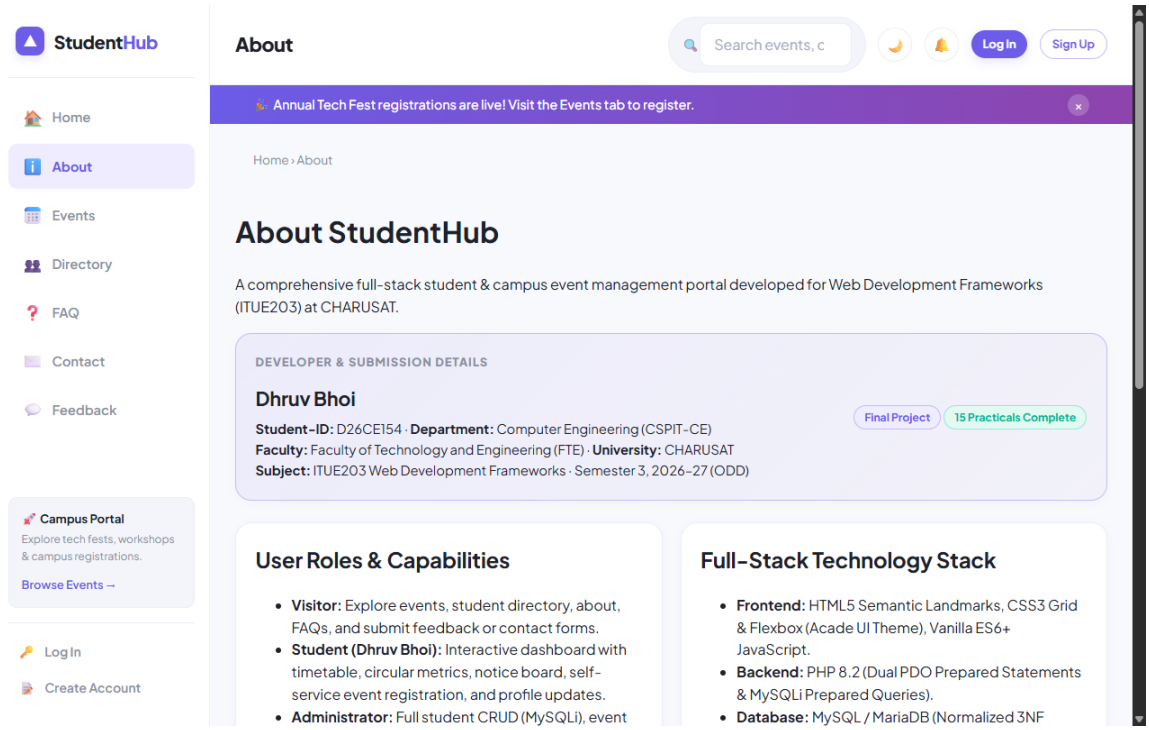


Figure: Modern Acade UI Shell & Student Developer Credentials Card (Dhruv Bhoi)

5. Key Questions & Analysis

Q: How does the layout adapt for mobile, tablet, and desktop screens?

Ans: Using a mobile-first responsive approach. Desktop screens display the persistent left navigation sidebar and multi-column grid cards. Tablet and mobile devices collapse the sidebar into an accessible slide-out drawer, transform grid layouts into single-column cards, and enable smooth touch scrolling.

Q: Which layout technique is used and why?

Ans: A hybrid approach: CSS Flexbox is used for 1-dimensional components (topbar headers, button groups, breadcrumbs, nav pills), while CSS Grid is used for 2-dimensional structures (dashboard timetable rows, event catalogs, stats cards, and multi-field forms).

Practical 3 Conclusion & Learning Outcomes

Engineered a modern, responsive user interface utilizing CSS Grid, Flexbox, media queries, and CSS variables. Redesigned the portal to match the Acade SaaS dashboard theme, featuring timetable schedule cards, circular SVG progress rings, a notice feed, and mobile off-canvas drawer navigation.